A-share cross-sectional stock selection

Four fixed methods | Top 100 | 20-session targets | 2 Jan 2018 – 29 Apr 2026

2021–2025 RIDGE NET
195.66%

2026 RIDGE NET
8.80%

2026 RIDGE NET EXCESS
+0.56 pp

Method	2021–2025 net	2026 net	2026 excess (pp)
Fundamental	139.90%	3.53%	-4.25
Combined	145.46%	9.73%	+1.33
Ridge	195.66%	8.80%	+0.56
LightGBM	178.42%	5.56%	-2.40
Benchmark	76.60%	8.18%	+0.00

Ridge adds net return; the latest-year excess remains modest.

Matched costs and historical pool. Absolute returns compound; excess sums daily differences.

Economic mechanisms behind the candidate inputs

Fourteen candidates define the common pool; fixed annual selections determine actual model inputs

Combine complementary hypotheses, then test their realised behaviour.

PROFITABILITY & CASH

ROE and ROA dynamics
Revenue-growth dynamics
ROA level
Operating cash / revenue

F1 · F2 · F3 · F8 · F9

VALUATION & LEVERAGE

Signed book / market
Signed TTM earnings yield
Debt / equity dynamics
Low debt / assets

F5 · F6 · F7 · F10

PRICE & TRADING

20- and 60-day momentum
126-to-20-day momentum
Return volatility
Average turnover

T1 · T2 · T3 · T5 · T6

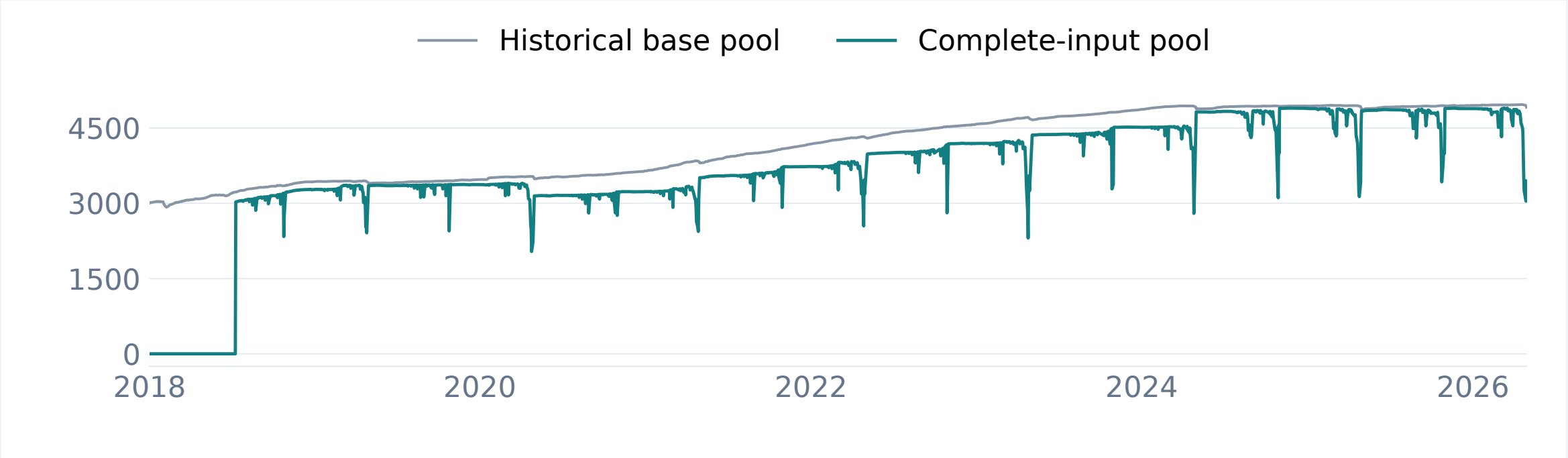
Signs and usefulness can vary; no economic story guarantees alpha.

Matched costs and historical pool. Absolute returns compound; excess sums daily differences.

Historical coverage defines the opportunity set

2 Jan 2018 – 29 Apr 2026 | 2,018 sessions | 6,000 historical slots

16 saved columns · 14 finite candidates · First signal: 11 Jul 2018



All methods share one historical pool; unavailable inputs stay missing.

Report period and both announcement dates are screened. Unrevised vendor vintages cannot be certified.

Matched costs and historical pool. Absolute returns compound; excess sums daily differences.

A fixed model and processing pipeline

Predict raw 20-session returns; update the scores every market day

Prepared inputs → **Within-date ranks** → **Fixed annual fits** → **Industry + log size** → **Causal EMA and rerank**

Method	Score or fixed model	EMA sessions
Fundamental	Signed financial ranks	20
Combined	Equal financial / technical families	40
Ridge	alpha = 10,000	20
LightGBM	depth 3; 7 leaves; 56 trees	40

Only mature earlier labels enter annual fits; parameters stay fixed.

Matched costs and historical pool. Absolute returns compound; excess sums daily differences.

Implementation and the matched benchmark

Daily signals and valuation; continuous twenty-session target refreshes

Trade at the next open; keep the actual cash and blocked orders.

PORTFOLIO & TIMING

Equal-weight Top 100
Next-opening execution
T+1 and price limits
Blocked-order retries

MATCHED BENCHMARK

Same eligible stock pool
Same twenty-day targets
Same restrictions / costs
CSI500 is auxiliary

TRADING FRICTIONS

Commission: 3 bp / side
Slippage: 5 bp / side
Sell stamp: 10 → 5 bp
Change: 28 Aug 2023

Twelve sourced corporate events use explicit claims and settlement dates.

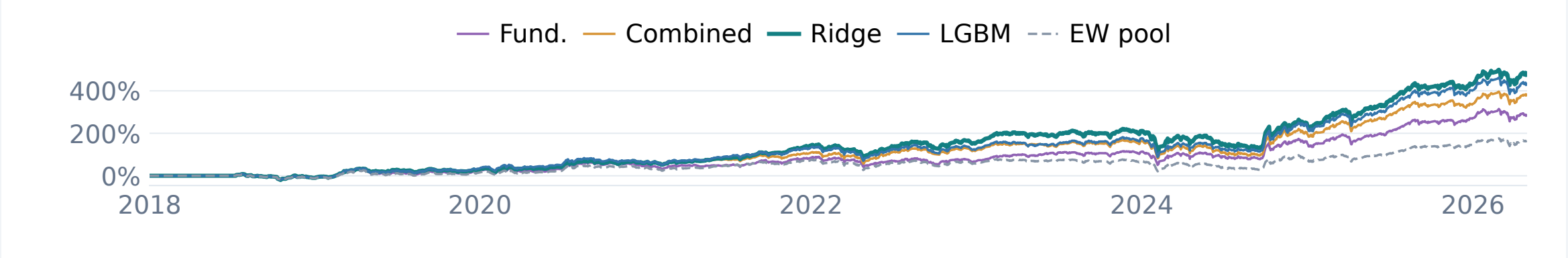
Synthetic adjusted units approximate dividends; exact receipt taxes, board lots and market impact are omitted.

Matched costs and historical pool. Absolute returns compound; excess sums daily differences.

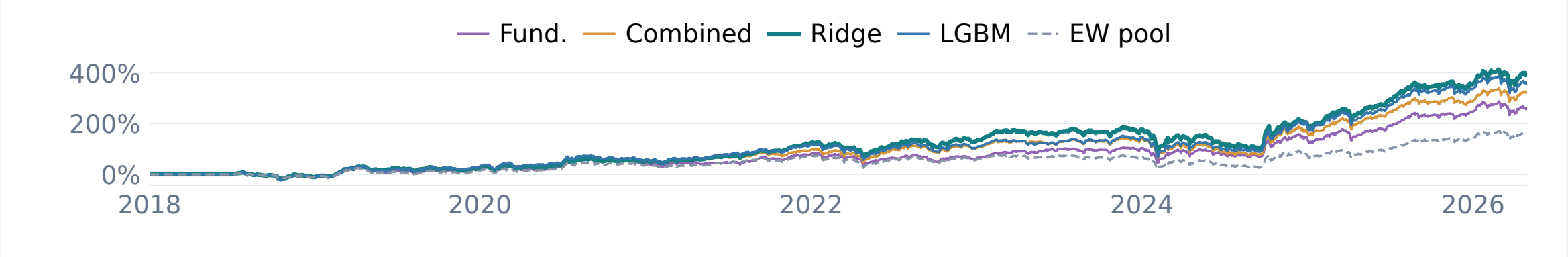
Gross and net returns over the complete history

2 Jan 2018 – 29 Apr 2026

GROSS ABSOLUTE RETURN



NET ABSOLUTE RETURN



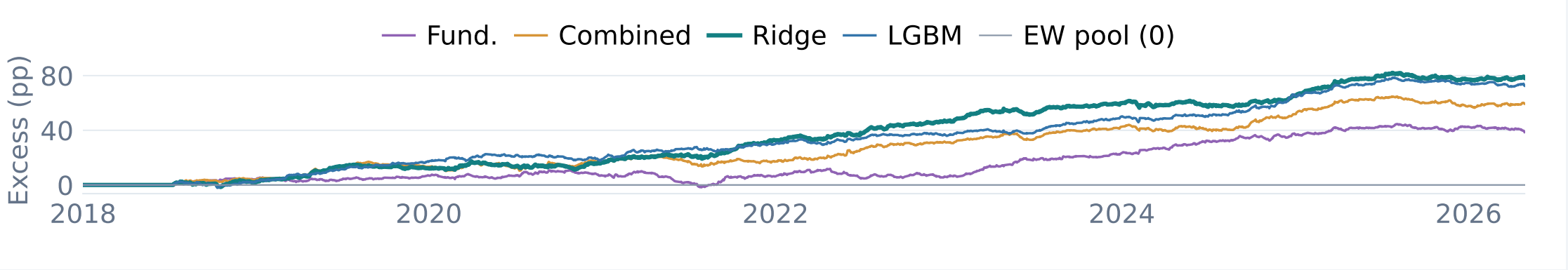
Absolute wealth compounds daily returns; net books pay their own costs.

Matched costs and historical pool. Absolute returns compound; excess sums daily differences.

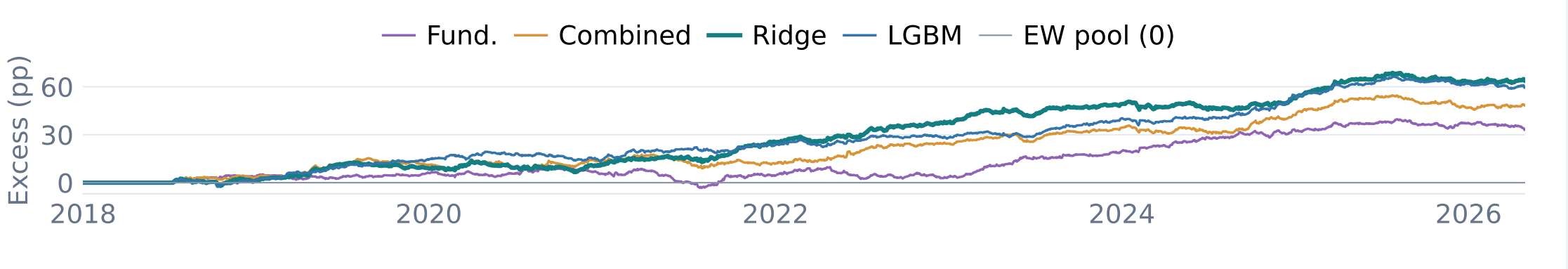
Gross and net excess over the same-cost benchmark

2 Jan 2018 – 29 Apr 2026

GROSS CUMULATIVE DAILY EXCESS (pp)



NET CUMULATIVE DAILY EXCESS (pp)



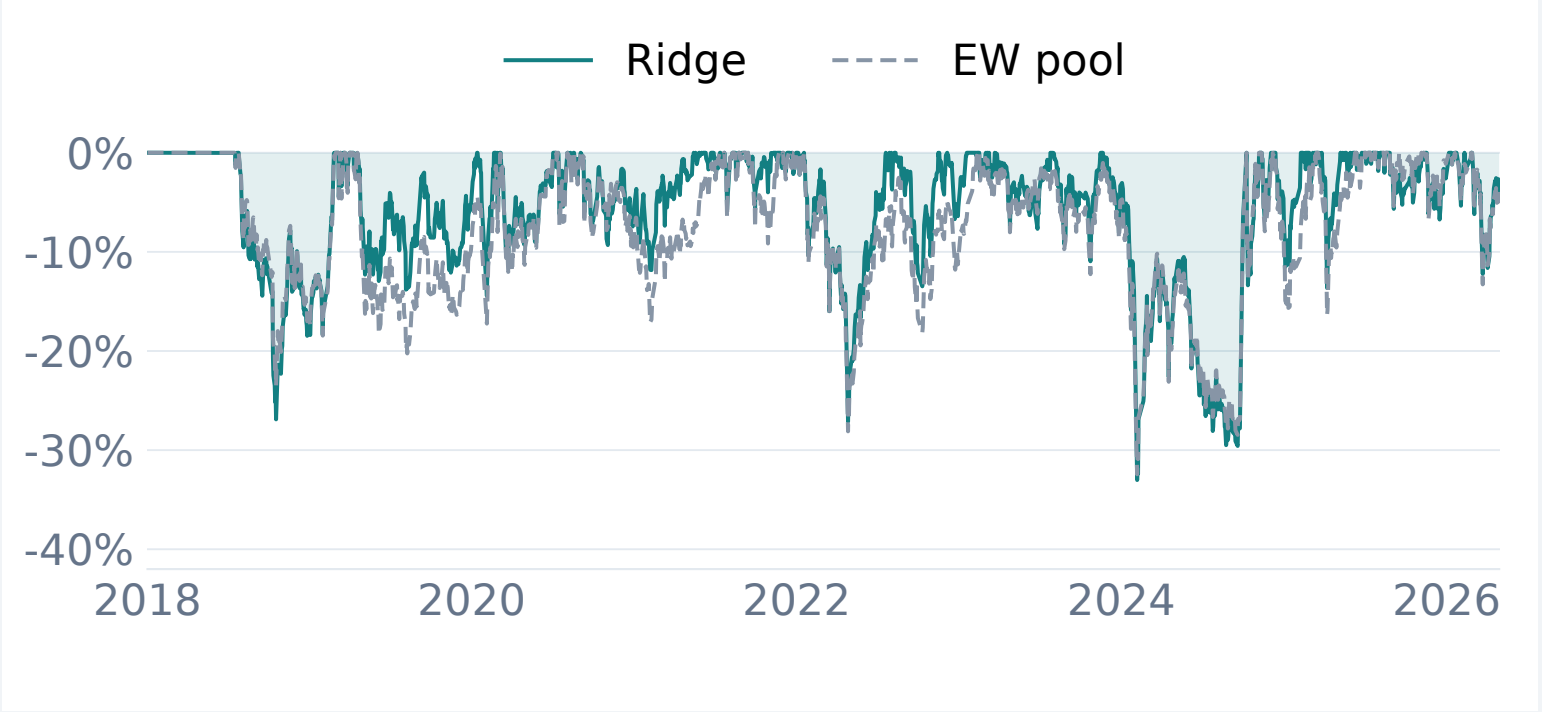
Excess sums daily return differences in percentage points.

Matched costs and historical pool. Absolute returns compound; excess sums daily differences.

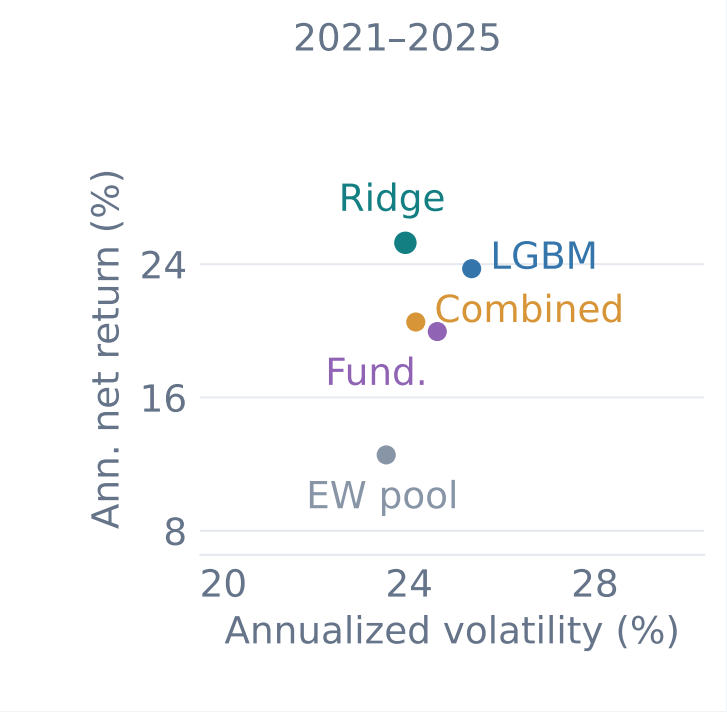
Higher return still comes with drawdown risk

Net portfolios | 2 Jan 2018 – 29 Apr 2026

PEAK-TO-TROUGH DRAWDOWN



RETURN AND VOLATILITY



Ridge worst drawdown: -33.0%; benchmark: -32.4%.

Matched costs and historical pool. Absolute returns compound; excess sums daily differences.

Excess varies across months and calendar years

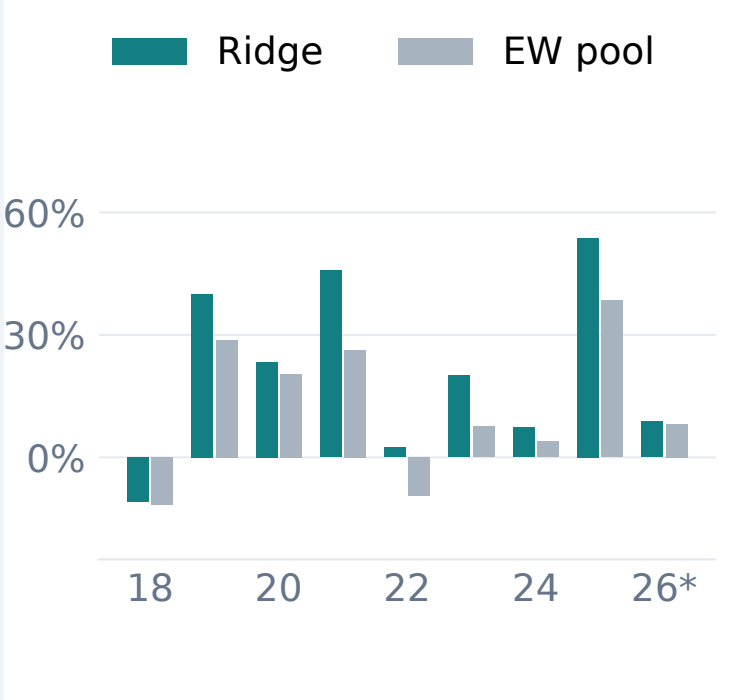
Monthly daily-excess sums and annual compounded net returns | 2026 through 29 Apr 2026

MONTHLY NET EXCESS · pp

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
2018	0.0	0.0	0.0	0.0	0.0	0.0	+2.2	-1.2	-0.6	-0.1	+1.9	-1.2
2019	+1.7	+0.7	+1.1	+3.0	+1.4	+2.6	+1.0	-1.2	-0.6	-0.1	-0.5	-1.0
2020	+0.1	+0.1	+4.5	-1.7	-1.6	-1.1	-0.7	+0.8	+0.5	-2.3	+2.5	+1.0
2021	+2.7	+1.3	-1.0	+1.4	-0.3	+0.3	-2.8	+2.9	+3.1	+4.0	0.0	+2.5
2022	+2.1	+0.4	-2.0	+1.9	+1.1	+0.4	+3.8	+2.4	+0.1	+0.6	+0.3	+1.5
2023	+2.1	+3.1	+1.6	+0.6	-1.1	-2.0	+4.7	+0.2	+0.1	+0.2	+1.4	+0.4
2024	+0.5	-1.6	-0.7	+2.3	+0.2	-3.7	+0.7	-0.1	+2.5	+0.2	+0.8	+2.8
2025	+3.7	+2.4	+3.2	+2.5	+0.1	+1.5	+1.5	-0.8	-2.6	+0.1	-0.4	-0.9
2026	+0.2	-0.3	+0.1	+0.5	—	—	—	—	—	—	—	—

-5 0 5 Monthly daily-excess sum (pp)

CALENDAR-YEAR NET RETURN



2026 is incomplete; future months are blank rather than zero.

Latest-year net excess: +0.56 pp; individual months can lag.

Matched costs and historical pool. Absolute returns compound; excess sums daily differences.

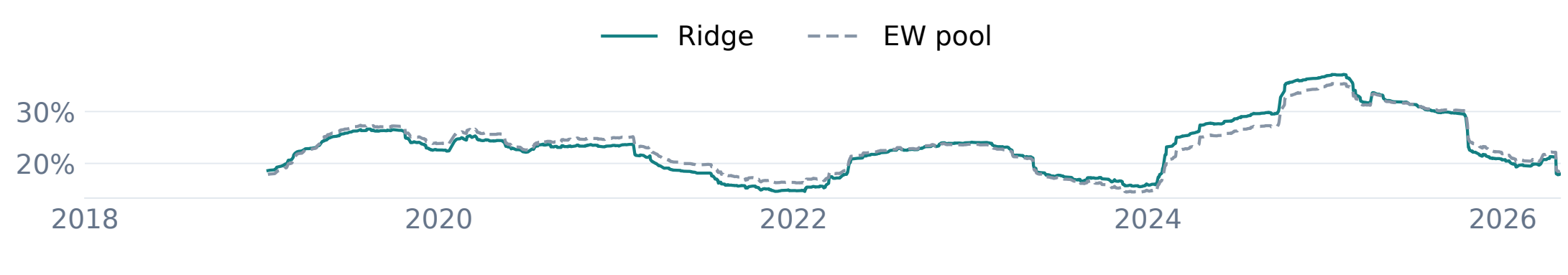
Rolling windows show changes in return and risk

252-session windows | 2 Jan 2018 – 29 Apr 2026

252-SESSION NET EXCESS (pp)



252-SESSION ANNUALIZED NET VOLATILITY (%)



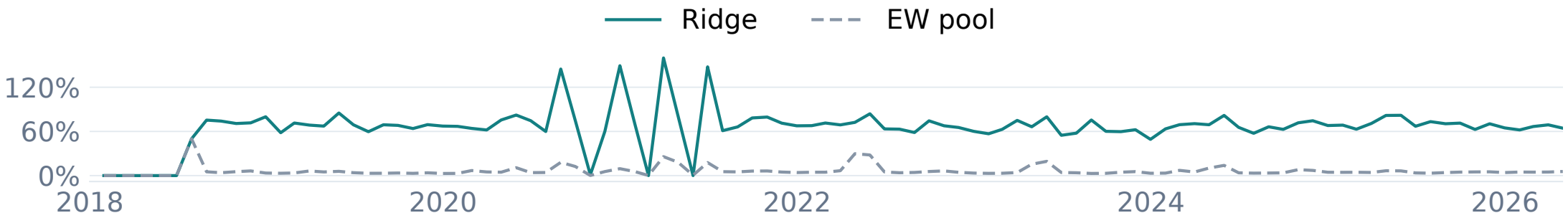
Relative performance and volatility change through time.

Matched costs and historical pool. Absolute returns compound; excess sums daily differences.

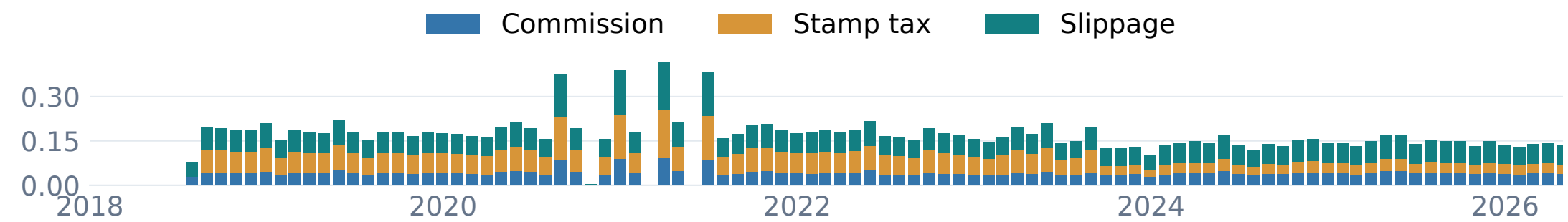
Turnover and trading costs remain material

2 Jan 2018 – 29 Apr 2026

MONTHLY TURNOVER



COST COMPONENTS · SUM OF DAILY OPENING-NAV RATIOS, pp



Costs alter cash, holdings and future returns in the net ledger.

Matched costs and historical pool. Absolute returns compound; excess sums daily differences.

Signal diagnostics reveal overlap and instability

Used-input dependence and cross-sectional prediction quality are different questions

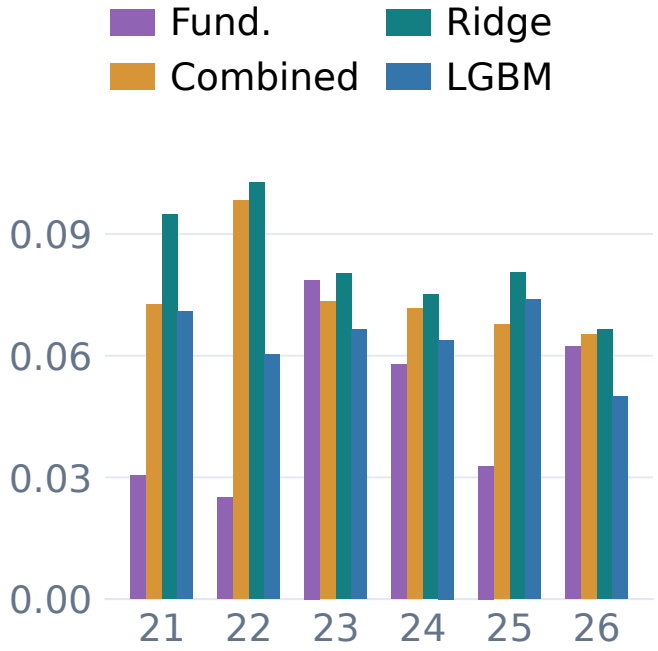
12 USED INPUTS · CORRELATION 2021–2025

	F1	F2	F3	F5	T1	T2	T3	T5	F6	F7	F8	F9
F1	+1.0											
F2	+0.1	+1.0										
F3	0.0	+0.2	+1.0									
F5	0.0	-0.1	0.0	+1.0								
T1	0.0	0.0	0.0	0.0	+1.0							
T2	+0.1	0.0	0.0	0.0	+0.5	+1.0						
T3	0.0	0.0	0.0	+0.1	+0.2	+0.3	+1.0					
T5	0.0	0.0	0.0	0.0	+0.1	+0.3	+0.7	+1.0				
F6	-0.1	0.0	0.0	-0.1	-0.1	-0.1	-0.4	-0.3	+1.0			
F7	0.0	-0.1	0.0	-0.1	0.0	0.0	-0.3	-0.3	+0.3	+1.0		
F8	+0.1	+0.1	0.0	-0.1	0.0	0.0	-0.1	-0.1	-0.2	+0.6	+1.0	
F9	+0.1	0.0	0.0	-0.1	0.0	0.0	-0.1	-0.1	0.0	+0.2	+0.4	+1.0

-1 0 1

Mean within-date Spearman

ANNUAL RANK IC



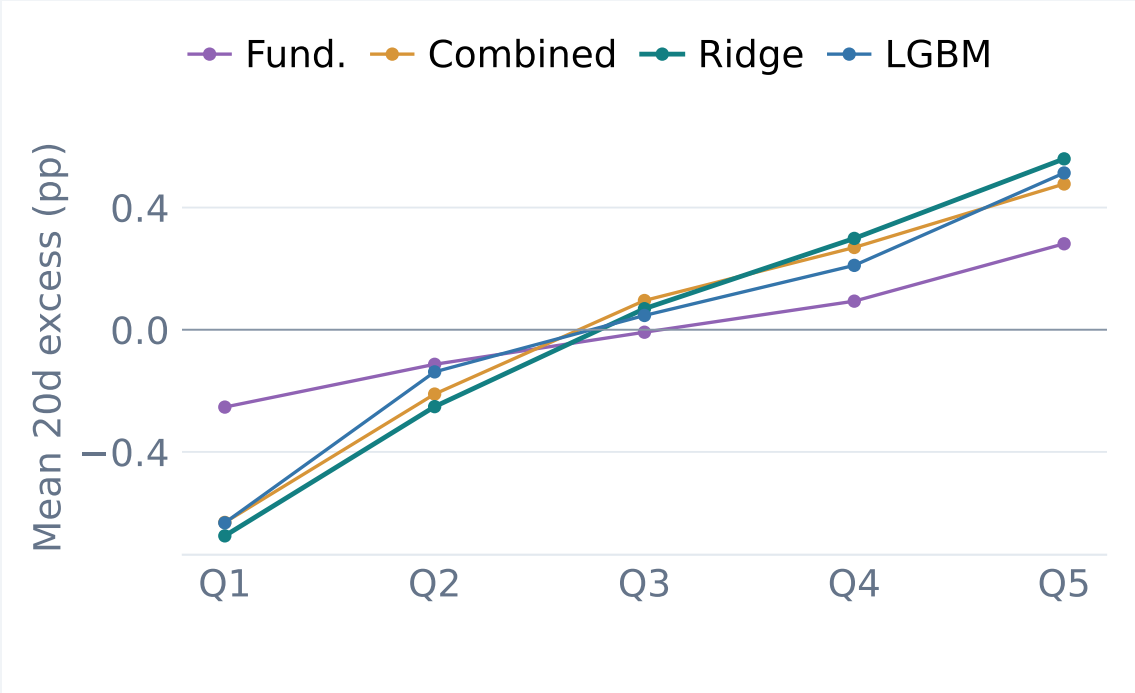
Signal correlation and IC do not establish a tradable net return.

Matched costs and historical pool. Absolute returns compound; excess sums daily differences.

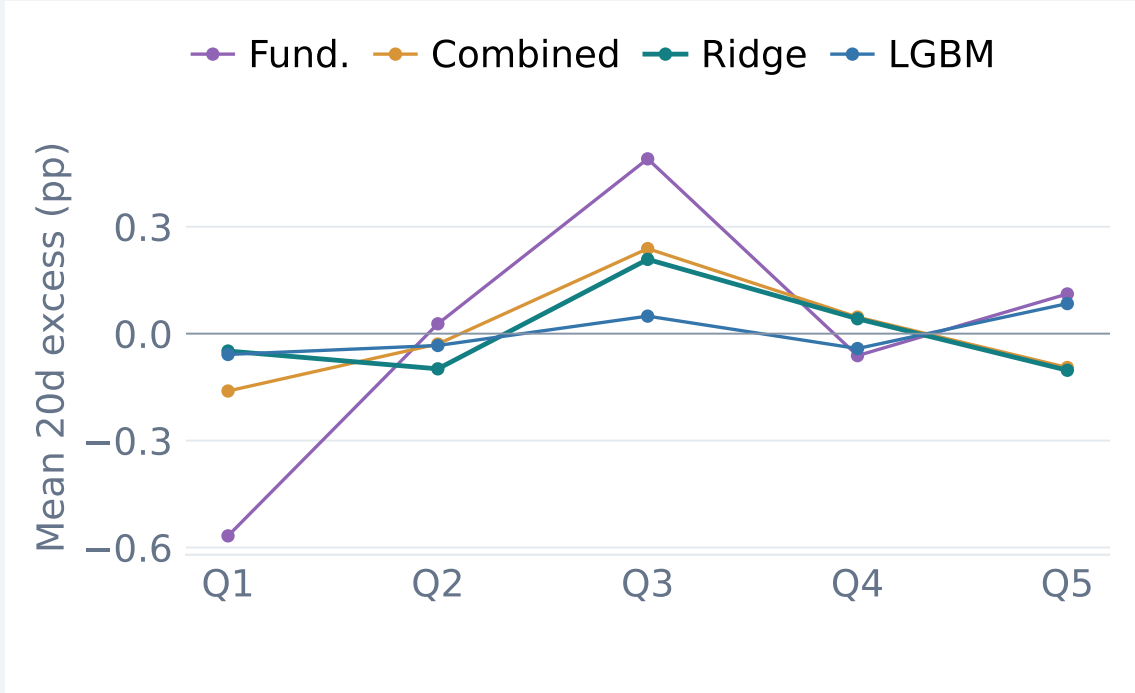
Do the highest scores identify stronger return cohorts?

Mean 20-session same-pool excess (pp), before costs | overlapping cohorts

2021–2025



2026 · MATURE LABELS ONLY



Higher scores need not produce a monotone return pattern in every period.

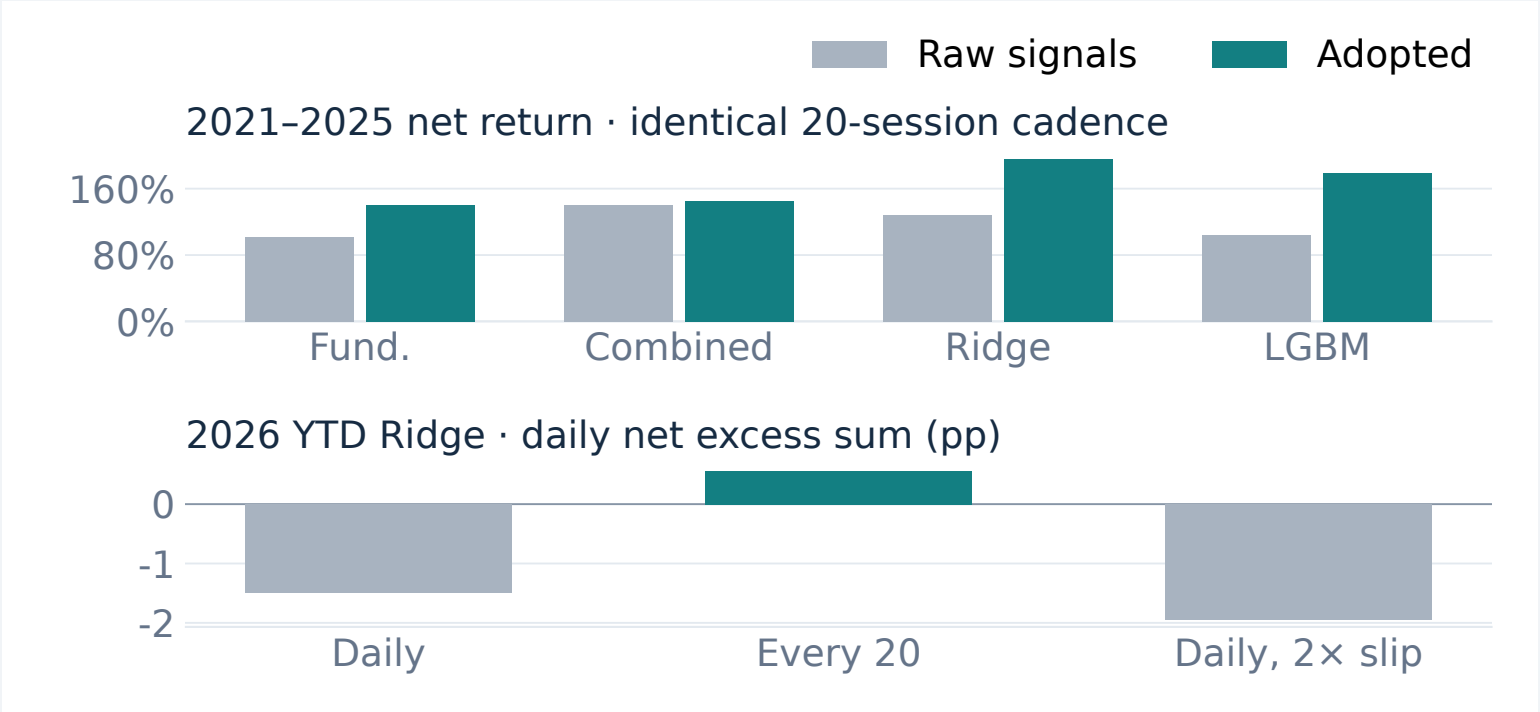
Group 1 = low, group 5 = high. Last mature signal: 30 Mar 2026.

Matched costs and historical pool. Absolute returns compound; excess sums daily differences.

Controls expose the effect of implementation choices

Same opportunity set and stated costs; retain weaker comparisons

RAW / ADOPTED AND DAILY / TWENTY-SESSION CONTROLS



2026 THROUGH 29 APR 2026

Method	Net %	Excess pp
Fundamental	3.53	-4.25
Combined	9.73	+1.33
Ridge	8.80	+0.56
LightGBM	5.56	-2.40
Benchmark	8.18	+0.00

Excess is a daily sum versus the matching-cost benchmark.

Implementation choices affect both realised return and costs.

Matched costs and historical pool. Absolute returns compound; excess sums daily differences.

Conclusions and the limits of the evidence

Actual historical data through 29 Apr 2026; no promise of future performance

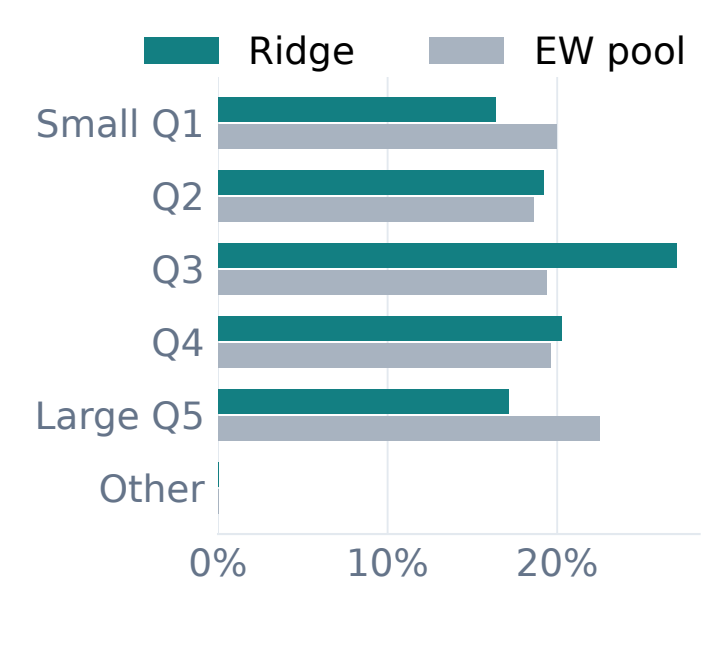
1. Ridge beats the matched benchmark over 2021–2025.

2. The latest-year net excess is +0.56 pp.

3. Score residualization does not make holdings style neutral.

4. Vendor revisions and execution assumptions remain limits.

ACTUAL FINAL CAPITAL WEIGHTS



Q3: Ridge 27.0%; benchmark 19.4%.

Judge return, risk and implementation together.

Matched costs and historical pool. Absolute returns compound; excess sums daily differences.